Homework #2

Math 104: Intro to Analysis

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Contents

1 Problem 1

Let (X, d) be a metric space. A subset $E \subset X$ is called *dense* in E if, for any open $O \subset X$, we have $O \cap E \neq \varnothing$. A metric space (X, d) is called *separable* if it has a countable dense subset.

(a) Show that $\mathbb{Q}$ is dense in $\mathbb{R}$, and conclude that $\mathbb{R}$ is separable.

(b) Show that a subset E is dense in X if and only if $\overline{E} = X$.

(c) By following these steps, prove that any compact metric space (X, d) is separable:

  (i) Using problem 5(c) on homework 2, show that for each $n \in \mathbb{N}$, the following is true: there is a finite set $S_n \subset X$ so that every point of X is within distance $\frac{1}{n}$ of some point of S_n.

  (ii) Let $S = \bigcup_{n=1}^{\infty} S_n$. Show that S is countable.

  (iii) Show that every point of X is either in S or a limit point of S. Conclude that S is dense in X, and hence that X is separable.

1.1 Solution (a)

1.2 Solution (b)

1.3 Solution (c)

2 Problem 2

Using **only** the definitions of lim, lim sup, and lim inf, prove the following statements.

(a) $\lim_{n \to \infty} \left[\sqrt{n+1} - \sqrt{n}\right] = 0.$ *Hint:* Use difference of squares.

(b) $\limsup_{n \to \infty} (-2)^n = \infty.$

(c) $\limsup_{n \to \infty} \left[-\sqrt{n}\right] = -\infty.$

(d) $\liminf_{n \to \infty} \left[\frac{1}{n} + (1 + (-1)^n)n\right] = 0.$

2.1 Solution (a)

2.2 Solution (b)

2.3 Solution (c)

2.4 Solution (d)

3 Problem 3

Let $\{s_n\}$ be an arbitrary sequence in $\mathbb{R}$.

(a) Let $s \in \mathbb{R}$. Show that $\limsup_{n\to\infty} s_n = s$ if and only if the following two conditions hold:

- For all $\epsilon > 0$, there is $N \in \mathbb{N}$ so that $s_n < s + \epsilon$ for all $n \geq N$.
- For all $\epsilon > 0$ and for all $N \in \mathbb{N}$, there exists $n \geq N$ so that $s_n > s - \epsilon$.

(b) Show that some subsequence of $\{s_n\}$ converges (in the sense of extended real numbers, if necessary) to $\limsup_{n\to\infty} s_n$ and some subsequence of $\{s_n\}$ converges to $\liminf_{n\to\infty} s_n$.

(c) Define

$$E = \{x \in [-\infty, \infty] : \text{ some subsequence } \{s_{n_k}\} \text{ of } \{s_n\} \text{ converges to } x\}.$$

Show that $\limsup_{n\to\infty} s_n = \sup E$ and $\liminf_{n\to\infty} s_n = \inf E$.

Hint: what happens to lim sup and lim inf on subsequences?

3.1 Solution (a)

3.2 Solution (b)

3.3 Solution (c)

4 Problem 4

Let $a > 1$. In this problem, we will define what it means to write a^r for a positive *irrational* number r. We first define exponentiation for positive *integer* exponents in the usual way, via repeated multiplication. We know this satisfies some properties, for example

$$a^{r+s} = a^r a^s \qquad (*)$$

$$(ab)^r = a^r b^r \qquad (\Delta)$$

for $a, b > 0$ and $r, s > 0$ integers. Now, we proved in class that for any positive integer q, there is a *unique* positive number whose qth power is a, which we denote $\sqrt[q]{a}$. For a positive rational number $r = \frac{p}{q}$, we define $a^r = \sqrt[q]{a^p}$.

(a) Show that this definition makes sense, i.e. is independent of the representation of r as a fraction. That is, show that if rational numbers $\frac{p}{q} = \frac{r}{s}$ then $\sqrt[q]{a^p} = \sqrt[s]{a^r}$. *Hint:* use lowest terms and the uniqueness of qth roots.

(b) Show that properties $(*)$ and (Δ) above still hold for positive rational numbers. *Hint:* find a common denominator and use uniqueness again.

(c) For positive rational $r < s < t$, show that $a^r < a^s$ and that

$$0 < a^s - a^r = a^r(a^{s-r} - 1).$$

Conclude that for any positive rational r, s and any rational t larger than both r and s, we have

$$|a^r - a^s| < a^t(a^{|r-s|} - 1).$$

(d) Now let r be a positive *irrational* number. We know there is a sequence of rational numbers q_k that converge to r (for example, the decimal expansion of r). We define

$$a^r = \lim_{k\to\infty} a^{q_k}.$$

Show that the sequence $\{a^{q_k}\}$ is Cauchy, and conclude that this limit exists.
Hint: Use part (c) and Rudin Theorem 3.20(d).

(e) Show that the choice of sequence $\{q_k\}$ is irrelevant; that is, show that if $\{p_k\}$ and $\{q_k\}$ are sequences of rational numbers converging to r, then

$$\lim_{k\to\infty} a^{p_k} = \lim_{k\to\infty} a^{q_k}.$$

(f) Using limit laws, prove that properties $(*)$ and (Δ) continue to hold for positive irrational exponents.

Remark: For $r < 0$ we then define $a^r = \frac{1}{a^{-r}}$, and for $0 < a < 1$ we define $a^r = \left(\frac{1}{a}\right)^{-r}$. It is then a matter of checking cases to verify that all the properties of exponentiation we want are indeed true.

4.1 Solution (a)

Proof. Let $x = \sqrt[q]{a^p}$ and $y = \sqrt[s]{a^r}$. This would mean that $x = a^{\frac{p}{q}}$ and $y = a^{\frac{r}{s}}$. Take x to the power of q and y to the power of s to find $x^q = a^p$ and $y^s = a^r$. OEΔ

4.2 Solution (b)

4.3 Solution (c)

4.4 Solution (d)

4.5 Solution (e)

4.6 Solution (f)